6b The terminal way (bash), harder but not tough FULL

24. Tools for learning and programming (very optional)

Quick version

No time to read it all? This is the short version. The full guide is below.

- **Nothing here is needed for the garden.** It runs without any of these tools.
- This page is for later, when you want to explore: an editor, a first programming language, a look at what the computer is doing.
- One table each for Windows, Ubuntu 24.04 and Raspberry Pi OS. Each table says what is already there.
- On Ubuntu and the Pi, most tools install with one line: `sudo apt install NAME`. `sudo apt remove NAME` takes it away again.
- Install from official pages only. One tool at a time. Never paste commands from strangers.

Nothing on this page is needed for the garden. The garden menu installs everything the garden uses. These tools are for when you want to explore, learn, or write a little code of your own. Skip this page as long as you like.

On Windows

Tool	What it is for	Already there?	How to get it
Windows Terminal	A window for typing commands, with tabs	Yes, in Windows 11	https://learn.microsoft.com/windows/terminal/
PowerShell	Windows' own command language (guide 20)	Yes, in Windows	https://learn.microsoft.com/powershell/
WSL	A real Ubuntu inside Windows	Yes, if the garden runs on this PC (the Windows menu switched it on)	https://learn.microsoft.com/windows/wsl/install
Visual Studio Code	An editor for code and text files	No	https://code.visualstudio.com/
Git	Keeps a history of changes to your files	No	https://git-scm.com/downloads/win
Python	A beginner-friendly programming language	No	https://www.python.org/downloads/windows/
Thonny	A simple editor for learning Python (Python comes with it)	No	https://thonny.org/
Geany	A light, fast editor	No	https://www.geany.org/

On Ubuntu 24.04

Tool	What it is for	Already there?	How to get it
nano	A simple editor inside the terminal (guide 22)	Yes	https://www.nano-editor.org/
Git	Keeps a history of changes to your files	Yes, the garden's install added it	https://git-scm.com/
Python (`python3`)	A beginner-friendly programming language	Yes, part of Ubuntu	https://www.python.org/
htop	Shows what the computer is doing, live (**q** leaves)	No	`sudo apt install htop` (https://htop.dev/)
tree	Shows folders as a tree	No	`sudo apt install tree` (https://packages.ubuntu.com/noble/tree)
Thonny	A simple editor for learning Python	No	`sudo apt install thonny` (https://thonny.org/)
Geany	A light, fast editor	No	`sudo apt install geany` (https://www.geany.org/)
Visual Studio Code	An editor for code and text files	No	`sudo snap install --classic code` (https://code.visualstudio.com/docs/setup/linux)

On Raspberry Pi OS (64-bit, with desktop)

Tool	What it is for	Already there?	How to get it
nano	A simple editor inside the terminal (guide 22)	Yes	https://www.nano-editor.org/
Git	Keeps a history of changes to your files	Yes	https://git-scm.com/
Python (`python3`)	A beginner-friendly programming language	Yes	https://www.raspberrypi.com/documentation/computers/os.html
Thonny	A simple editor for learning Python	Yes	https://thonny.org/
Geany	A light, fast editor	Yes	https://www.geany.org/
htop	Shows what the computer is doing, live (**q** leaves)	Yes	https://htop.dev/
tree	Shows folders as a tree	No	`sudo apt install tree` (https://packages.debian.org/trixie/tree)
Visual Studio Code	An editor for code and text files	No	`sudo apt install code` (https://code.visualstudio.com/docs/setup/raspberry-pi)

For later

Tool	What it is for	How to get it
Rust (with rustup)	The language the garden website is written in. The garden builds the website without it (inside Docker)	https://rustup.rs/
Android Studio	Builds the garden's phone app (guide 17)	Windows and Ubuntu: https://developer.android.com/studio. Not on the Raspberry Pi: Google's page says Linux computers with ARM processors are not supported yet

Rust on Windows: download and run `rustup-init.exe` from https://rustup.rs/.

Rust on Ubuntu or the Pi: the one line from https://rustup.rs/:

```
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
```

How to install a program from the terminal on Ubuntu or the Pi

```
sudo apt install NAME
```

Put the program's name in place of `NAME`, for example `sudo apt install tree`. It asks for your password (nothing shows while you type). If it asks "Do you want to continue?", type `y` and press Enter.

To remove it again:

```
sudo apt remove NAME
```

Ubuntu's official page: https://ubuntu.com/server/docs/how-to/software/package-management/

Good habits

- **Install from official pages only**, like the ones on this page.

- **One tool at a time.** Try it, then decide on the next.
- **Never paste commands from strangers** (a forum, a video, a message).
- Try things in your own playground folder (guide 22), not in the garden folder.

Next: 22 Bash, step by step